

Introduction to Probability

Reasoning Under Uncertainty

Artificial Intelligence

From Logical to Probabilistic Intelligence I

In the previous lecture, we considered systems that reason using **logical propositions**. Such systems are powerful when we can establish facts reliably.

The problem

Real intelligent systems rarely have complete and perfectly reliable information.

Examples:

- Sensors produce noisy measurements.
- Observations may be incomplete.
- Different explanations may be possible.
- The world itself may be unpredictable.
- Our models of the world may be imperfect.

From Logical to Probabilistic Intelligence II

We therefore need a way to reason when we cannot say simply

true or false.

The Assumptions of Logical Reasoning I

A purely logic-based system implicitly benefits from assumptions such as:

- assertions are accurate;
- observations are complete and error-free;
- alternative conclusions can be treated as equally viable;
- inference preserves truth.

These assumptions are often unrealistic in complex environments.

Key idea

The problem is not that logical reasoning is wrong.

The problem is that **logic alone does not represent uncertainty**.

Why Uncertainty is Fundamental to AI I

Consider an autonomous vehicle.

It may observe:

- a collection of pixels;
- noisy sensor readings;
- an object that may be a pedestrian;
- uncertain estimates of position and velocity.

The system might therefore have beliefs such as

$$P(\text{pedestrian}) = 0.92$$

rather than the logical assertion

pedestrian = true.

Why Uncertainty is Fundamental to AI II

Probability gives us a language

Probability allows an AI system to represent and manipulate **degrees of belief** in propositions and events.

Probability is More Than Randomness I

It is tempting to associate probability only with **random processes**:

- coin tosses;
- dice;
- games of chance.

In AI, probability has a broader role.

We use probability to represent uncertainty arising from:

Source of uncertainty	Example
Noisy observations	Sensor measurement
Incomplete information	Missing features
Unpredictable environment	Weather
Uncertain model	Future behaviour
Ambiguous evidence	Medical diagnosis

Probability as Degree of Belief I

Suppose an AI system is considering the proposition

$$A = \text{"It will rain tomorrow."}$$

It might assign

$$P(A) = 0.8.$$

This does *not* mean that the proposition is "80% true."

Rather, it represents the system's current **degree of belief** in A .

As new information arrives, the belief can change:

$$P(A) = 0.8 \quad \longrightarrow \quad P(A \mid E) = 0.95.$$

Here E represents newly observed evidence.

Probability as Degree of Belief II

Central idea

Probabilistic reasoning allows beliefs to be **updated as evidence becomes available**.

A Simple AI Example: Diagnosis I

Suppose an AI system is diagnosing a machine fault.

Let

F = “the cooling system has failed”

and

E = “the temperature is unusually high”.

Before observing the temperature, the system may have

$$P(F) = 0.05.$$

After observing the high temperature, it may have

$$P(F \mid E) = 0.65.$$

A Simple AI Example: Diagnosis II

The important question is therefore not simply

Is F true?

but

How strongly should the evidence change our belief in F ?

Probability and Intelligent Behaviour I

Probability becomes useful because intelligent behaviour often requires three steps:

Represent uncertainty $\longrightarrow$ Update beliefs $\longrightarrow$ Make decisions

For example:

- ① Estimate the probability that an object is a pedestrian.
- ② Update that belief using additional sensor observations.
- ③ Choose an appropriate action.

This gives us a probabilistic foundation for many AI systems, including:

- diagnosis;
- perception;
- prediction;
- decision making;
- planning under uncertainty.

What Will Probability Give Us? I

By the end of this lecture we want to be able to answer questions such as:

- How likely is an event?
- How should we represent uncertainty?
- How should beliefs change when evidence arrives?
- How can we combine different sources of evidence?
- When can two uncertain quantities be treated as independent?

The fundamental mathematical objects will be:

$$P(A)$$

and

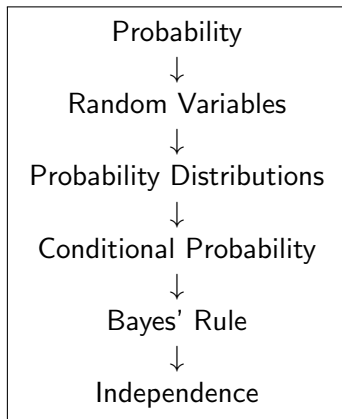
$$P(A \mid B).$$

The first represents a belief before conditioning on B .

The second represents a belief after taking B into account.

The Road Ahead I

We will build the ideas in stages.



The Road Ahead II

Then we will ask a more difficult question:

Next lecture

How can we represent a probability distribution over **many interacting variables** without explicitly storing every possible combination?

This leads naturally to

Bayesian Networks.

From Logic to Probability I

We can now contrast the two viewpoints.

Logical reasoning	Probabilistic reasoning
Is A true?	How likely is A ?
A or $\neg A$	$0 \leq P(A) \leq 1$
Certain facts	Degrees of belief
Truth-preserving inference	Belief updating
Complete information is ideal	Uncertainty is explicit

These are not competing forms of intelligence.

AI perspective

Logic and probability provide complementary tools for reasoning about different kinds of knowledge and uncertainty.

From Variables to Random Variables I

In logic, a variable represents an unknown object or value.

In probabilistic AI, we often want to represent an **uncertain quantity**.

We therefore introduce a **random variable**.

Random variable

A random variable is a variable whose value depends on the outcome of an uncertain process or observation.

For example:

$$Weather \in \{\text{sunny, rain, cloudy, snow}\}.$$

The important point is that we do not know in advance which value *Weather* will take.

Random Variables in AI I

Random variables allow us to describe uncertain aspects of a world.

Random variable	Possible values
Weather	sunny, rain, cloudy, snow
TrafficLight	red, amber, green
Pedestrian	true, false
Diagnosis	flu, COVID, other
BatteryLevel	low, medium, high

A random variable does *not* necessarily represent something physically random.

AI interpretation

It represents a quantity whose value is uncertain **to the agent**.

Values, Assignments and States I

Suppose our system has three random variables:

Weather, *Traffic*, *Pedestrian*.

An assignment such as

Weather = rain, *Traffic* = heavy, *Pedestrian* = true

specifies one possible state of the world.

State

A **state** is a complete assignment of values to all random variables in the model.

For example,

$s = (\text{rain}, \text{heavy}, \text{true}).$

A state is therefore one possible “world” that the AI system is considering.

Worlds and Events I

Suppose our model contains the Boolean variable

$$Pedestrian \in \{\text{true}, \text{false}\}.$$

There are two possible states:

State	<i>Pedestrian</i>
s_1	<i>true</i>
s_2	<i>false</i>

The proposition

$$Pedestrian = \text{true}$$

corresponds to the set of states in which the proposition is true.

Event

An **event** is a set of possible states satisfying some condition or proposition.

Thus, an event may contain one state or many states.

A Simple Example: Four Possible Worlds I

Consider two Boolean variables:

Rain, *Traffic*.

There are four possible states:

	<i>Rain</i>	<i>Traffic</i>
s_1	<i>false</i>	<i>light</i>
s_2	<i>false</i>	<i>heavy</i>
s_3	<i>true</i>	<i>light</i>
s_4	<i>true</i>	<i>heavy</i>

The event

Rain = *true*

A Simple Example: Four Possible Worlds II

contains two states:

$$\{s_3, s_4\}.$$

The event

$$Rain = true \wedge Traffic = heavy$$

contains only

$$\{s_4\}.$$

Key idea

A proposition identifies a collection of possible worlds.

Probability Assigns Weight to Events I

Probability tells us how much belief we assign to an event.

For an event A ,

$$0 \leq P(A) \leq 1.$$

For example,

$$P(Rain) = 0.30$$

means that, given what the system currently knows, it assigns probability 0.30 to the event that it will rain.

Similarly,

$$P(Rain \wedge Traffic = heavy) = 0.20.$$

The probability is therefore attached to a **proposition/event**, not directly to a truth value.

The Three Axioms of Probability I

A valid probability model satisfies three fundamental axioms.

Axiom 1: Non-negativity

$$P(A) \geq 0$$

for every event A .

Axiom 2: Normalisation

$$P(\Omega) = 1,$$

where Ω is the set of all possible states.

The Three Axioms of Probability II

Axiom 3: Additivity

If A and B cannot both occur,

$$P(A \vee B) = P(A) + P(B).$$

These simple rules impose strong constraints on every probability model.

What Do the Axioms Mean? I

The axioms give probability an important interpretation.

Probability 0

$$P(A) = 0$$

means that A is assigned no probability.

Probability 1

$$P(A) = 1$$

means that A is certain under the model.

Everything else lies between these extremes:

$$0 < P(A) < 1.$$

What Do the Axioms Mean? II

Important

Probability is not a second kind of truth.
It is a quantitative representation of uncertainty.

Complementary Events I

For any event A , either A occurs or $\neg A$ occurs.

These events are mutually exclusive and collectively exhaustive.

Therefore,

$$P(A) + P(\neg A) = 1.$$

Equivalently,

$$P(\neg A) = 1 - P(A).$$

For example, if

$$P(\textit{Pedestrian}) = 0.92,$$

then

$$P(\neg Pedestrian) = 1 - 0.92 = 0.08.$$

A First Worked Example I

An autonomous vehicle's perception system assigns

$$P(Pedestrian) = 0.92.$$

What is the probability that the detected object is **not** a pedestrian?

Using the complement rule,

$$P(\neg Pedestrian) = 1 - P(Pedestrian).$$

Therefore,

$$\begin{aligned} P(\neg Pedestrian) &= 1 - 0.92 \\ &= 0.08. \end{aligned}$$

A First Worked Example II

Answer

The system assigns probability

0.08

to the object not being a pedestrian.

A Probability Distribution I

A **probability distribution** specifies how probability is distributed across the possible values of a random variable.

Consider

$$Weather \in \{\text{sunny, rain, cloudy, snow}\}.$$

Suppose:

$$P(Weather = \text{sunny}) = 0.70$$

$$P(Weather = \text{rain}) = 0.20$$

$$P(Weather = \text{cloudy}) = 0.08$$

$$P(Weather = \text{snow}) = 0.02.$$

These probabilities sum to one:

A Probability Distribution II

$$0.70 + 0.20 + 0.08 + 0.02 = 1.$$

Checking a Probability Distribution I

A discrete probability distribution must satisfy two conditions.

- 1 Every probability is non-negative:

$$P(X = x) \geq 0.$$

- 2 The probabilities of all possible values sum to one:

$$\sum_x P(X = x) = 1.$$

Checking a Probability Distribution II

For the weather example:

$$0.70 + 0.20 + 0.08 + 0.02 = 1.$$

Therefore this is a valid probability distribution.

AI interpretation

The distribution represents the system's uncertainty about the value of X .

Worked Example: Is This a Valid Model? I

Suppose an object detector produces the following distribution:

Object	P
<i>pedestrian</i>	0.65
<i>bicycle</i>	0.20
<i>car</i>	0.10
<i>other</i>	0.05

Is this a valid probability distribution?

Check:

$$0.65 + 0.20 + 0.10 + 0.05 = 1.$$

Every probability is also non-negative.

Worked Example: Is This a Valid Model? II

Conclusion

Yes. This is a valid probability distribution.

Probability of a Compound Event I

A probability distribution can answer questions about **compound events**.
Suppose

$$Weather \in \{\text{sunny, rain, cloudy, snow}\}$$

with

$$P(\text{sunny}) = 0.70$$

$$P(\text{rain}) = 0.20$$

$$P(\text{cloudy}) = 0.08$$

$$P(\text{snow}) = 0.02.$$

What is the probability of *not* having sunny weather?

$$P(\neg \text{sunny}) = P(\text{rain}) + P(\text{cloudy}) + P(\text{snow}).$$

Probability of a Compound Event II

Thus,

$$P(\neg \textit{sunny}) = 0.20 + 0.08 + 0.02 = 0.30.$$

Equivalently,

$$P(\neg \textit{sunny}) = 1 - P(\textit{sunny}) = 0.30.$$

Mutually Exclusive Events I

Two events are **mutually exclusive** if they cannot both occur.
For example, in a single observation:

Weather = sunny

and

Weather = rain

cannot both be true.

Therefore,

$$P(\textit{sunny} \wedge \textit{rain}) = 0.$$

For mutually exclusive events,

Mutually Exclusive Events II

$$P(A \vee B) = P(A) + P(B).$$

This is one of the most useful consequences of the axioms.

Worked Example: Combining Evidence Categories I

An AI classifier assigns:

$$P(\textit{pedestrian}) = 0.60$$

$$P(\textit{cyclist}) = 0.25$$

$$P(\textit{vehicle}) = 0.15.$$

Assume these categories are mutually exclusive.

What is the probability that the object is **not a vehicle**?

$$\begin{aligned} P(\neg \textit{vehicle}) &= P(\textit{pedestrian} \vee \textit{cyclist}) \\ &= P(\textit{pedestrian}) + P(\textit{cyclist}) \\ &= 0.60 + 0.25 \\ &= 0.85. \end{aligned}$$

Therefore,

Worked Example: Combining Evidence Categories II

$$P(\neg vehicle) = 0.85 .$$

From One Variable to Several I

So far we have considered one random variable at a time.
But AI systems usually reason about many quantities simultaneously.
For example:

Weather, Traffic, Pedestrian.

Now we may want to ask:

$$P(\textit{Weather} = \textit{rain} \wedge \textit{Traffic} = \textit{heavy}).$$

Or:

$$P(\textit{Pedestrian} = \textit{true} \wedge \textit{Traffic} = \textit{heavy}).$$

These are examples of **joint probabilities**.

Joint probability

A joint probability describes the probability that several events occur together.

A Small Joint Distribution I

Consider two Boolean variables:

Rain, *TrafficHeavy*.

A joint distribution might be:

	<i>TrafficHeavy</i>	$\neg$ <i>TrafficHeavy</i>
<i>Rain</i>	0.20	0.10
$\neg$ <i>Rain</i>	0.15	0.55

Every cell represents a complete assignment.

For example,

$$P(Rain \wedge TrafficHeavy) = 0.20.$$

The four cells describe all possible states of the two variables.

A Small Joint Distribution II

Check:

$$0.20 + 0.10 + 0.15 + 0.55 = 1.$$

The Full Joint Distribution I

For random variables

$$X_1, X_2, \dots, X_n,$$

the **full joint distribution** specifies

$$P(X_1 = x_1, \dots, X_n = x_n)$$

for every possible complete assignment.

In principle, this gives us everything we need to know about the probabilistic model.

But there is a problem

The number of possible assignments grows rapidly as the number of variables increases.

For n Boolean variables there are

The Full Joint Distribution II

$$2^n$$

possible states.

The Representation Problem I

Suppose we have:

Number of Boolean variables	Possible states
5	32
10	1,024
20	1,048,576
30	1,073,741,824
100	2^{100}

A full joint distribution therefore becomes impractical very quickly.

Key AI challenge

How can we represent and reason with a probability distribution over many variables without explicitly storing every possible state?

This question will motivate Bayesian networks.

Question

An AI system has three mutually exclusive hypotheses:

$$H_1, \quad H_2, \quad H_3.$$

It assigns

$$P(H_1) = 0.50, \quad P(H_2) = 0.30.$$

- 1 What must $P(H_3)$ be?
- 2 What is $P(\neg H_1)$?
- 3 Why must the three probabilities sum to one?

Think–Pair–Share:

Can you solve all three questions using only the axioms of probability?

Formative Assessment: Solution I

Since the hypotheses are mutually exclusive and exhaustive,

$$P(H_1) + P(H_2) + P(H_3) = 1.$$

Therefore,

$$P(H_3) = 1 - 0.50 - 0.30 = \boxed{0.20}.$$

For the complement of H_1 ,

$$P(\neg H_1) = 1 - P(H_1) = \boxed{0.50}.$$

The probabilities sum to one because exactly one of the three exhaustive hypotheses must hold.

Key Takeaways I

We have introduced the basic language of probabilistic AI.

- A **random variable** represents an uncertain quantity.
- A **state** is a complete assignment of values.
- An **event** is a set of states satisfying a proposition.
- A **probability distribution** assigns probabilities to possible values or states.
- Probability satisfies three fundamental axioms.
- Complementary and mutually exclusive events give useful rules for combining probabilities.
- A **joint distribution** represents several variables simultaneously.

The next question is:

How should beliefs change when we observe evidence?

From Probability to Belief Updating I

In Part II, we assigned probabilities to possible events.
But intelligent systems receive new information.

Suppose:

$$P(\text{Rain}) = 0.30.$$

Now suppose we observe that the sky is heavily overcast.
Our belief about rain should change.

Key question

How should we represent the probability of an event **given some evidence**?

This leads to

conditional probability.

Conditional Probability I

The conditional probability of A given B is written

$$P(A \mid B).$$

It represents the probability of A when we know that B has occurred.
Formally,

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)}$$

provided that

$$P(B) > 0.$$

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should be read as

“given that”.

What Does Conditioning Do? I

Consider the joint distribution:

	<i>TrafficHeavy</i>	$\neg$ <i>TrafficHeavy</i>
<i>Rain</i>	0.20	0.10
$\neg$ <i>Rain</i>	0.15	0.55

Suppose we learn that

$$\textit{TrafficHeavy} = \textit{true}.$$

We can now ignore the states in which traffic is not heavy.

The relevant probability mass is

$$0.20 + 0.15 = 0.35.$$

Therefore,

What Does Conditioning Do? II

$$P(Rain \mid TrafficHeavy) = \frac{0.20}{0.35} \approx 0.571.$$

Interpretation

Conditioning changes the probability distribution by restricting attention to worlds consistent with the evidence.

A Useful Way to Think About Conditioning I

Before observing B :

$$P(A)$$

represents our current belief about A .

After observing B :

$$P(A \mid B)$$

represents our updated belief.

Thus:

$$P(A) \longrightarrow P(A \mid B)$$

can be interpreted as

A Useful Way to Think About Conditioning II

prior belief $\longrightarrow$ updated belief.

This is one of the central operations in probabilistic AI.

The Product Rule I

Starting from the definition

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)},$$

multiply both sides by $P(B)$:

$$P(A \wedge B) = P(A \mid B)P(B).$$

Equivalently,

$$P(A \wedge B) = P(B \mid A)P(A).$$

These are two forms of the

product rule.

Why the Product Rule Matters I

The product rule gives us two ways to calculate a joint probability:

$$P(A \wedge B) = P(A \mid B)P(B)$$

or

$$P(A \wedge B) = P(B \mid A)P(A).$$

For example,

$$P(\text{Rain} \wedge \text{TrafficHeavy})$$

can be calculated as

$$P(\text{Rain} \mid \text{TrafficHeavy})P(\text{TrafficHeavy})$$

or as

Why the Product Rule Matters II

$$P(\text{TrafficHeavy} \mid \text{Rain})P(\text{Rain}).$$

Important

The two expressions are equal, but the conditional probabilities are generally **not** equal.

A Common Source of Confusion I

In general,

$$P(A \mid B) \neq P(B \mid A).$$

For example:

$$P(\text{Disease} \mid \text{Positive test})$$

is the probability that a patient has the disease given a positive test.

This is different from

$$P(\text{Positive test} \mid \text{Disease}),$$

which is the probability that the test is positive given that the patient has the disease.

A Common Source of Confusion II

AI lesson

The direction of conditioning matters.

The Law of Total Probability I

Suppose an event B can occur under several mutually exclusive and exhaustive possibilities:

$$A_1, A_2, \dots, A_n.$$

Then

$$P(B) = \sum_i P(B \mid A_i)P(A_i).$$

This is the

law of total probability.

It says:

To calculate the probability of an event, consider all the ways in which it can occur.

Example: Total Probability I

Suppose an AI system receives data from two types of sensor.

$$P(S_1) = 0.6, \quad P(S_2) = 0.4.$$

The probability of a false alarm is

$$P(F | S_1) = 0.05$$

and

$$P(F | S_2) = 0.20.$$

What is the overall probability of a false alarm?

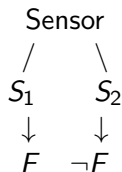
Using total probability:

Example: Total Probability II

$$\begin{aligned}P(F) &= P(F \mid S_1)P(S_1) + P(F \mid S_2)P(S_2) \\&= (0.05)(0.6) + (0.20)(0.4) \\&= 0.03 + 0.08 \\&= \boxed{0.11}.\end{aligned}$$

Probability Trees I

The law of total probability can be visualised using a probability tree.



Each complete path represents a joint event.

For example:

$$P(S_1 \wedge F) = P(S_1)P(F \mid S_1).$$

Thus:

$$P(F) = P(S_1 \wedge F) + P(S_2 \wedge F).$$

Key idea

Probability trees provide an intuitive representation of the product rule and total probability.

Bayes' Rule I

We have two forms of the product rule:

$$P(A \wedge B) = P(A \mid B)P(B)$$

and

$$P(A \wedge B) = P(B \mid A)P(A).$$

Since both describe the same joint probability,

$$P(A \mid B)P(B) = P(B \mid A)P(A).$$

Therefore,

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Bayes' Rule II

This is

Bayes' rule.

The Four Parts of Bayes' Rule I

Bayes' rule is often written

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

The terms have standard names:

Quantity	Name
$P(A)$	Prior
$P(B A)$	Likelihood
$P(B)$	Evidence / marginal likelihood
$P(A B)$	Posterior

The Four Parts of Bayes' Rule II

The conceptual transformation is:

$$\text{Prior} + \text{Evidence} \longrightarrow \text{Posterior.}$$

Bayes' rule gives us a general mechanism for updating beliefs.

$$P(A \mid B) \propto P(B \mid A)P(A)$$

where the proportionality constant ensures that the resulting probabilities sum to one.
In words:

- ① Start with a prior belief about A .
- ② Ask how likely the evidence B would be if A were true.
- ③ Combine the prior and the likelihood.
- ④ Normalise to obtain the posterior.

Worked Example: Medical Test I

Suppose a disease affects

1%

of a population:

$$P(D) = 0.01.$$

A diagnostic test has sensitivity

$$P(+ \mid D) = 0.90.$$

For people without the disease, the false-positive rate is

$$P(+ \mid \neg D) = 0.20.$$

A patient receives a positive result.

Worked Example: Medical Test II

What is

$$P(D \mid +)?$$

Medical Test: Step 1 I

Bayes' rule gives

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+)}.$$

We know

$$P(+ \mid D) = 0.90$$

and

$$P(D) = 0.01.$$

Therefore the numerator is

$$P(+ \mid D)P(D) = (0.90)(0.01) = 0.009.$$

We still need

$$P(+).$$

Medical Test: Step 2 I

The positive result can occur in two ways:

Disease and positive: $D \wedge +$

No disease and positive: $\neg D \wedge +$.

Therefore, by total probability,

$$\begin{aligned}P(+)&= P(+ \mid D)P(D) + P(+ \mid \neg D)P(\neg D) \\&= (0.90)(0.01) + (0.20)(0.99) \\&= 0.009 + 0.198 \\&= 0.207.\end{aligned}$$

Medical Test: Step 3 I

We can now calculate the posterior:

$$\begin{aligned}P(D \mid +) &= \frac{P(+ \mid D)P(D)}{P(+)} \\&= \frac{0.009}{0.207} \\&\approx 0.043.\end{aligned}$$

Therefore,

$$P(D \mid +) \approx 4.3\%.$$

Surprising result

Even after a positive test, the probability of disease is only about 4.3%.

The reason is the relatively low prior probability of the disease and the substantial false-positive rate.

Why the Result is Counterintuitive I

The test is quite good at detecting the disease:

$$P(+ \mid D) = 0.90.$$

But this is *not* the quantity we wanted.

We wanted

$$P(D \mid +).$$

These are different:

$$P(+ \mid D) \neq P(D \mid +).$$

The distinction is crucial in AI applications such as:

- diagnosis;

Why the Result is Counterintuitive II

- fraud detection;
- anomaly detection;
- spam filtering;
- fault diagnosis.

Bayesian reasoning

Evidence must be interpreted relative to the prior distribution of possible explanations.

Normalisation I

There is another useful way to view Bayes' rule.

Suppose we calculate an unnormalised score:

$$P(B | A)P(A).$$

For the two possible hypotheses A and $\neg A$:

$$\begin{aligned}q_A &= P(B | A)P(A), \\ q_{\neg A} &= P(B | \neg A)P(\neg A).\end{aligned}$$

We then normalise:

$$P(A | B) = \frac{q_A}{q_A + q_{\neg A}}.$$

Thus:

$$\text{posterior} = \frac{\text{unnormalised belief}}{\text{sum of all unnormalised beliefs}}.$$

A General Bayesian Update I

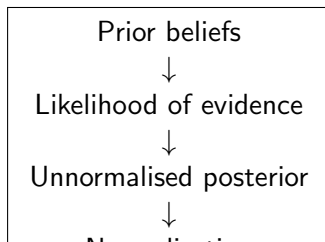
For hypotheses

$$H_1, \dots, H_n$$

and evidence E :

$$P(H_i | E) = \frac{P(E | H_i)P(H_i)}{\sum_j P(E | H_j)P(H_j)}.$$

This gives a general recipe:



A General Bayesian Update II

This is one of the fundamental computational patterns in probabilistic AI.

A Second Example: Cavity and Toothache I

Consider three Boolean variables:

Cavity, *Toothache*, *Catch*.

Suppose the joint distribution is:

	Toothache		$\neg$ Toothache	
	Catch	$\neg$ Catch	Catch	$\neg$ Catch
Cavity	0.108	0.012	0.072	0.008
$\neg$ Cavity	0.016	0.064	0.144	0.576

This is a full joint distribution: every cell represents a complete assignment.

Cavity Example: Conditioning I

Suppose we observe

$$\textit{Toothache} = \textit{true}.$$

We want

$$P(\textit{Cavity} \mid \textit{Toothache}).$$

From the definition of conditional probability,

$$P(\textit{Cavity} \mid \textit{Toothache}) = \frac{P(\textit{Cavity} \wedge \textit{Toothache})}{P(\textit{Toothache})}.$$

The numerator is

$$0.108 + 0.012 = 0.120.$$

Cavity Example: Conditioning II

The denominator is

$$0.108 + 0.012 + 0.016 + 0.064 = 0.200.$$

Therefore,

$$P(\text{Cavity} \mid \text{Toothache}) = \frac{0.120}{0.200} = 0.60.$$

Cavity Example: What Happened? I

Before observing the toothache, the joint distribution gives

$$P(\text{Cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = 0.20.$$

After observing the toothache:

$$P(\text{Cavity} \mid \text{Toothache}) = 0.60.$$

Thus the evidence has changed our belief:

0.20	→	0.60.
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This is Bayesian updating in action.

Cavity Example: What Happened? II

Important

The evidence does not “prove” that the patient has a cavity.
It changes the probability assigned to that hypothesis.

What Bayes' Rule Really Does I

Bayes' rule is sometimes presented as a formula to memorise:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

But its deeper significance is conceptual.

What did I believe before?



What Bayes' Rule Really Does II

How expected is this evidence under each hypothesis?



What should I believe now?

This is the basic pattern of probabilistic reasoning.

A Crucial Distinction I

Consider:

$$P(D \mid +)$$

and

$$P(+ \mid D).$$

The first asks:

How likely is the disease given a positive test?

The second asks:

How likely is a positive test given the disease?

A Crucial Distinction II

Bayes' rule connects them:

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+)}.$$

Do not confuse likelihood with posterior

$P(+ \mid D)$ is a likelihood.

$P(D \mid +)$ is a posterior belief.

Formative Question I

Suppose

$$P(A) = 0.2, \quad P(B \mid A) = 0.8, \quad P(B \mid \neg A) = 0.1.$$

Which of the following is closest to

$$P(A \mid B)?$$

- ① 0.16
- ② 0.20
- ③ 0.67
- ④ 0.80

Formative Question II

Hint:

First calculate

$$P(B)$$

using the law of total probability.

Formative Question: Solution I

First,

$$P(\neg A) = 1 - 0.2 = 0.8.$$

Using total probability,

$$\begin{aligned} P(B) &= P(B \mid A)P(A) + P(B \mid \neg A)P(\neg A) \\ &= (0.8)(0.2) + (0.1)(0.8) \\ &= 0.16 + 0.08 \\ &= 0.24. \end{aligned}$$

Therefore, by Bayes' rule,

Formative Question: Solution II

$$\begin{aligned}P(A \mid B) &= \frac{P(B \mid A)P(A)}{P(B)} \\&= \frac{0.16}{0.24} \\&\approx 0.67.\end{aligned}$$

Hence the correct answer is

(c) 0.67.

Summary of Part III I

We have developed a chain of ideas:

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)}$$

gives conditional probability.

From this:

$$P(A \wedge B) = P(A \mid B)P(B)$$

gives the product rule.

Summing over alternative explanations gives:

$$P(B) = \sum_i P(B \mid A_i)P(A_i)$$

Summary of Part III II

the law of total probability.

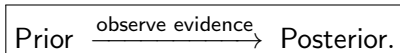
Combining the product rule in two directions gives:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Bayes' rule.

The Probabilistic AI Pattern I

The central computational pattern is:



More explicitly:

$$P(H) + P(E | H) \longrightarrow P(H | E).$$

This allows an AI system to:

- maintain uncertainty about possible explanations;
- incorporate new observations;
- revise its beliefs;
- compare competing hypotheses.

Looking Ahead I

We now have the basic machinery for reasoning with uncertainty.

But there is still a major problem.

Suppose we have many variables:

$$X_1, X_2, \dots, X_n.$$

A full joint distribution can represent everything, but may require an exponential number of entries.

We therefore want to exploit structure such as

independence

and, more importantly,

conditional independence.

Looking Ahead II

These ideas will lead us to

Bayesian Networks.